Venkata Sai

Senior Software Engineer, Trust Emerging Defense
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Professional Experience

Senior Software Engineer *Jul 2025 – Present*
Persistent Systems | Trust & Safety Platform

- Designed and implemented **scalable fraud detection** systems using **Java** and **Apache Flink**, processing 100K real-time transactions/second for enhanced user **safety**.
- Led development of **machine learning pipelines** with **Python** and **Kubeflow** for anomaly detection, reducing false positives in **risk investigation** by 15% through improved model accuracy.
- Architected and deployed **fault-tolerant distributed systems** leveraging **AWS SQS** and **Apache Kafka** for secure cross-service data exchange, ensuring 99.99% uptime for sensitive data.
- Collaborated with **cross-functional partners** to define product strategy for new **privacy** features on a **communication platform**, improving data protection for 2M daily active users.
- Developed **real-time risk investigation tools** using **React** and **Spring Boot**, enabling security teams to triage critical **unattended risks** 50% faster, enhancing platform integrity.
- Optimized existing **batch** processing jobs for **fraud mitigation** using **Spark** on **AWS EMR**, decreasing processing time by 40% for 500GB daily data volumes.
- Engineered a **marketplace of products** for identifying malicious activity with **Generative AI** models via **OpenAI API**, improving **fraud detection** rates by 25% for new threat vectors.

Software Engineer *Jul 2022 – Jun 2025*
Endava | Enterprise Risk & Compliance Platform

- Developed **RESTful APIs** in **Java Spring Boot** for a compliance data ingestion service, handling 1M requests daily while maintaining strict data **privacy** protocols.
- Implemented **data pipelines** using **Apache Kafka** and **Spark Streaming** for **real-time** transaction monitoring, supporting **fraud detection** across diverse data sources.
- Contributed to design and development of **scalable distributed systems** in **Python** for a financial risk assessment platform, reducing processing latency by 20% for 5M customer records.
- Integrated **machine learning models** for identifying suspicious user behavior using **scikit-learn** and **TensorFlow**, improving **fraud mitigation** efficiency by 18% with predictive analytics.
- Collaborated with **data scientists** to refine data schemas and optimize query performance in **PostgreSQL** for **risk investigation tools**, accelerating reporting by 30%.
- Engineered **security features** like data encryption and access management with **JWT** within a **service-oriented architecture**, bolstering user data **safety** for enterprise clients.
- Developed **automated tests** and **CI/CD pipelines** using **Jenkins** and **GitLab CI** to ensure code quality and rapid deployment of **privacy**-focused services, improving release frequency by 25%.
- Implemented a **batch** processing system for historical data analysis using **AWS S3** and **Glue**, enabling comprehensive **fraud detection** insights across 1TB of archived data.

Software Engineer *Jul 2020 – Aug 2022*
Nagarro | E-commerce Payment Gateway

- Developed and maintained **Java Spring Boot** microservices for a payment gateway, ensuring high availability and processing 500K transactions daily with improved **safety**.
- Implemented secure API integrations with third-party payment processors using **OAuth** and **REST**, enhancing transaction **security** and reliability by 15% through robust authentication.
- Optimized **MySQL** database queries and schemas for a high-volume e-commerce platform, reducing average transaction response times by 25% for a better user experience.
- Contributed to the design of **scalable distributed systems** handling user authentication and authorization using **Spring Security**, supporting growth to 1M registered users without downtime.
- Collaborated with **cross-functional teams** including product and QA to deliver new features for a digital wallet application, increasing user engagement by 10% through seamless integration.

Skills

Languages: Java, Python, SQL
Frameworks & Tools: Spring Boot, Apache Flink, Kubeflow, React, Spark, Spring Security, scikit-learn, TensorFlow, Apache Kafka, Spark Streaming, JUnit, Mockito, Jenkins, GitLab CI, OAuth, Generative AI, OpenAI API
Cloud & Infra: AWS SQS, AWS EMR, AWS S3, AWS Glue, PostgreSQL, MySQL
Concepts: Scalable Distributed Systems, Fault-Tolerant Systems, Service-Oriented Architecture, Architectural Patterns, Fraud Detection, Fraud Mitigation, Data Privacy, User Safety, Real-time Processing, Batch Processing, Risk Investigation, CI/CD, Security Features, Data Pipelines, Microservices, API Design

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India